Machine Learning Project Documentation

1. Project Title

Interactive BMI Prediction System using Linear Regression

2. Problem Statement

Body Mass Index (BMI) is an important indicator used to evaluate a person's health based on body weight relative to height. Predicting BMI can help understand health patterns and support better awareness of personal health metrics.

The goal of this project is to build a Machine Learning model that predicts BMI based on age using Linear Regression and deploy the model as an interactive web application using Streamlit.

3. Objective

- Build a regression model to predict BMI.
- Understand the relationship between age and BMI.
- Visualize the relationship between variables using graphs.
- Deploy the model as an interactive web application using Streamlit.

4. Dataset Information

The dataset contains information about individuals including age and BMI values. This dataset is used to train a machine learning model to predict BMI.

Example Features:
Age – Age of the individual
BMI – Body Mass Index value

The dataset is stored in a file named 'bmi.csv'.

5. Tools & Technologies

- Python
- Jupyter Notebook
- Pandas
- Scikit-learn
- Matplotlib

- Seaborn
- Streamlit
- GitHub

6. Data Preprocessing

The following steps were performed during data preprocessing:

1. Load dataset using pandas
2. Explore dataset structure
3. Select relevant features
4. Split dataset into training and testing data

Example:

```
X = data[['age']]
y = data['bmi']
```

7. Model Building

This project uses the Linear Regression algorithm to predict BMI.

Linear Regression models the relationship between an independent variable (Age) and a dependent variable (BMI).

Formula:

$$y = mx + c$$

Where:

y = Predicted BMI
x = Age
m = Slope
c = Intercept

8. Model Evaluation

The model performance is evaluated using the following metrics:

- R^2 Score
- Mean Squared Error (MSE)

These metrics help determine how accurately the model predicts BMI values.

9. Deployment

The trained machine learning model is deployed using Streamlit to create an interactive web application. Users can enter their age and the model will predict their BMI value.

The application also displays dataset previews, model metrics, and visualization graphs.

10. Results

The model successfully predicts BMI values based on the input age. The Streamlit interface allows users to interact with the model and obtain predictions instantly.

11. GitHub Repository

Project code available at:
https://github.com/HarshithaSeggem/linearRegression-project

12. Streamlit Deployment

Live application can be deployed using Streamlit Cloud.
Example:
https://linearregression-project-sbnplqynaxp3gxdydpjp2x.streamlit.app/

13. Conclusion

This project demonstrates how machine learning techniques can be applied to predict BMI using Linear Regression. The project includes data visualization, model evaluation, and deployment using Streamlit.

14. Future Scope

- Use additional features such as gender, smoking status, and number of children
- Improve prediction accuracy with advanced models
- Enhance the user interface of the web application
- Deploy the project on cloud platforms for wider accessibility